

# NinaXander: Feasibility and Limits of Composing Frozen Language Models Across Architecture Families via a Shared Latent Space

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**Abstract:** In this paper we propose NinaXander, a series of composed language models obtained by connecting layers of frozen language models from different architecture families with a single trained shared-latent adapter. A composed model runs the first layers of one model, converts the resulting intermediate representation once with the adapter, and then runs the remaining layers of the other model. Once the adapter is trained, several composed models that connect at different layers are obtained without retraining. Using the recurrent RWKV-4-Raven-7B and the Transformer-based Tulu-Pythia-6.9b, abbreviated as RWKV and Pythia, this study examines whether frozen models from different families can be recombined post hoc. The composed models answered multiple-choice questions, and those whose generations we examined produced syntactically well-formed text. The configuration that combines the first 5 layers of Pythia with the remaining 27 layers of RWKV reduced the Transformer key-value (KV) cache by 84.4% with accuracy not significantly different from that of RWKV alone. In multiple-choice accuracy, however, no composed model matched the parent model Pythia, and language-modeling performance decreased sharply on WikiText, a corpus of Wikipedia articles outside the training domain. The correspondence between intermediate representations was also obtained in one favorable case, with a shared tokenizer, the same depth, and the same hidden width, and does not show that the models share a general semantic space.

## 1. Introduction

Pretraining a language model requires an enormous amount of computation, yet its result remains only as weights fixed to a particular architecture. The Transformer, which most current language models use, keeps the keys and values of past tokens as a key-value (KV) cache to compute self-attention, and the size of this cache grows in proportion to the context length. Existing ways of changing inference-time characteristics through the architecture, for example to reduce the KV cache, either pretrain a new hybrid model that incorporates recurrent layers or replace some layers of an existing model with a lighter mechanism and retrain it. Both approaches retrain weights; neither recombines trained blocks as frozen components.

In this paper, a family is a group of models that share the same kind of architecture, such as Transformers or RWKV (Receptance Weighted Key Value), which computes linear attention in a recurrent form and so runs in time linear in the sequence length. Studies that connect frozen language models to one another have been reported, but in these studies either both connected models are Transformers or the whole composed model is fine-tuned, and recombining

trained checkpoints across families as frozen components has not yet been examined.

This study examines whether the intermediate representations of frozen models from different families can be converted into each other by a small trained network, called the adapter below, so that the first layers of one model can be connected to the remaining layers of another. We call a model composed in this way a chimera model and the original models used for the composition parent models, and we name the series of chimera models built with a single adapter NinaXander<sup>\*1</sup>. Once the adapter is trained, several chimera models that connect at different layers are obtained without retraining. The parent models are RWKV-4-Raven-7B, a version of RWKV further trained to follow instructions, and the Transformer-based Tulu-Pythia-6.9b. The two models satisfy the favorable conditions of sharing a tokenizer, which splits text into token sequences, and of having the same hidden width and the same number of layers. Below, the two models are abbreviated as RWKV and Pythia, respectively.

The main purpose of this study is not to build a practical high-performance model but to clarify whether frozen

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<sup>\*1</sup> The name NinaXander comes from a chimera in Hiromu Arakawa's manga *Fullmetal Alchemist*. The composition works but falls short of what it was made from, which corresponds to the results of this study.

models from different families can be connected by converting their intermediate representations. In our experiments the chimera models answered multiple-choice questions, and those whose generations we examined produced syntactically well-formed text. We also obtained a configuration that reduces the Transformer layers to 5, decreasing the KV cache size by 84.4%, with accuracy not significantly different from that of RWKV alone. On the other hand, no configuration exceeded the stronger parent model, and RWKV alone, which has no KV cache, was significantly exceeded in accuracy by only one chimera model, on the science question set SciQ. The contributions of this study are the construction and evaluation of chimera models that connect frozen language models from different families in both directions with a single adapter; an analysis of the factors that limit the performance of the connection, through a comparison with reconstruction within the same model and an intervention inside the adapter; and measurements of accuracy, KV cache size, and performance on text from domains not used to train the adapter.

## 2. Background and Related Work

### 2.1 Residual Stream and Inference Cost

An autoregressive language model updates the representation at each token position, called the residual stream, layer by layer. Block  $\ell$ , counted from 0, of a model with hidden width  $d$  receives the input  $h^{\ell-1}$  and outputs  $h^\ell = h^{\ell-1} + f_\ell(h^{\ell-1})$ , where  $h^{-1}$  is the token embedding. The function  $f_\ell$  performs sequence mixing, which mixes information across tokens, and channel mixing, which transforms features within each token position. Even when two models have the same hidden width and depth, the  $h^\ell$  of one model lies in a coordinate system determined by that model's training, and passing it without conversion to block  $\ell+1$  of the other model does not match the distribution that the receiving weights expect. Whether two models can be connected at an intermediate layer therefore reduces to whether the residuals of the two families can be converted into each other.

The two families also differ in how their inference cost grows. The self-attention of Transformers weights tokens by their similarity normalized with the softmax function, and because it keeps the keys and values of all past tokens it needs a cache of size  $O(T)$  for a context length  $T$ . A layer that can run as linear attention or in an equivalent recurrent form keeps only a fixed-size state, which is  $O(1)$ . Which of the two connected models is placed on the input side changes how many Transformer layers remain and how much cache must be kept.

### 2.2 Model Stitching

Lenc and Vedaldi [1] introduced connecting layers of 2 frozen models through a trainable stitching layer as a way to measure the equivalence of representations. For two convolutional neural networks (CNNs), they built a “Franken-CNN” that connects the lower layers of one network to the

upper layers of the other through a linear stitching layer, and evaluated it with the loss of the original task. The pairs they studied include networks trained on different data, namely the large-scale image classification dataset ImageNet and the scene recognition dataset MIT Places built by the Massachusetts Institute of Technology (MIT). They also include pairs of CNNs with different architectures: the 8-layer AlexNet, the 16-layer VGG-16 proposed by the Visual Geometry Group of the University of Oxford, and the 50-layer ResNet-50 with residual connections. Bansal et al. [2] revisited stitching as a means of comparing representations. They stitched models of the same architecture that differ in random seed, amount of training, and width, and also stitched across different objectives, namely supervised learning and self-supervised learning, which trains on pseudo-labels constructed from the data itself. Stitching across different architectures, however, remained an open problem. Csiszárík et al. [3] matched the intermediate representations of two trained networks with a single affine stitching layer and proposed a framework that evaluates their agreement not only by similarity in representation space but also functionally, by the task performance of the stitched network. They further showed that existing measures of representational similarity do not necessarily reflect task performance. Following this position, we evaluate separately the degree to which representations agree, called alignment below, and the degree to which one representation can be recovered from the other, called decodability below and measured in this paper by the read-out accuracy (§3.5). All of these studies target image recognition models. This study differs in that it connects two different families, recurrent neural networks (RNNs) and Transformers, through a common latent space in which the representation is converted once at an intermediate layer.

### 2.3 Composing Frozen Large Language Models

Recent studies compose frozen large language models (LLMs) through trainable connection layers. StitchLLM [4] connects blocks of Transformers of different sizes through stitching layers and uses a lightweight routing mechanism to decide how much computation to allocate to each query. BTS (Branch-Train-Stitch) [5] places several frozen expert models, derived from the same seed model for specific domains, alongside the frozen seed model, which serves as the hub, and inserts Experts-into-Hub and Hub-into-Experts stitching layers alternately every few layers to exchange hidden states in both directions. CALM (Composition to Augment Language Models) [6] fuses the intermediate representations of two frozen Transformers through cross-attention, which uses the representation of one model as queries to attend to the representation of the other. Manticore [7] runs block sequences of a Transformer and of Mamba, a selective state-space model, in parallel with projections, and mixes their outputs with learned weights that are non-negative and sum to 1. This hybrid model is fine-tuned as a whole, and its embedding layer and its language-model head (LM

head), which outputs probabilities over the vocabulary, are newly initialized.

Other studies convert activations directly between frozen models. Moschella et al. [8] showed that relative representations, which re-express each representation by its similarity to a small set of reference samples called anchors, remove the effect of the transformation between independently trained latent spaces, so that models can be stitched without training a converter. Maiorca et al. [9] estimate a map between two latent spaces in closed form from a small number of semantic correspondences. As low-capacity direct maps corresponding to these methods, this study uses affine maps for comparison (§4.3). Because our affine maps are fitted directly with corresponding residuals as targets, the comparison is more favorable to them than to relative representations, which do not use paired data for training.

Model Alignment Search [10] maps activations in both directions between a frozen recurrent network and a frozen Transformer, but it is a method for measuring representational similarity and does not build a composed model. Chen et al. [11] map the residual streams of two frozen language models in both directions with a single affine layer, and evaluate, down to the cross-entropy computed from next-token prediction probabilities, composed models that connect the first layers of one model to the remaining layers of the other. Their pairs, however, are Transformers of one model series sharing a tokenizer; the future work they name is validation across model series, and they do not discuss different families.

The Bicameral Model [12], a preprint, couples two frozen language models through bidirectional conversion networks, but each model runs in parallel while executing all of its own layers. GoldFinch [13], also a preprint, stacks a Transformer on a structure that extends RWKV-6 (Finch), an improved version of RWKV-4, and compresses the KV cache by 756–2550 times, but it is trained from an initial state rather than combining pretrained parent models. Oozeer et al. [14] learn activation maps between frozen LLMs and transfer steering vectors that control tendencies of the output, mainly between smaller and larger models of one series, using them to remove backdoors planted so that specific inputs trigger undesired outputs and to disable the refusal of harmful instructions. To validate a map, they replace the activations of one target layer with mapped ones and generate text, and they use affine maps as a control, so their work is close to ours; each map, however, serves one layer pair, and both models are Transformers.

## 2.4 Representation Similarity

Standard measures of representational similarity between networks are centered kernel alignment (CKA) [15] and two extensions of canonical correlation analysis (CCA), singular vector CCA (SVCCA) and projection weighted CCA (PWCCA) [16], [17]. Of these, we use linear CKA, together with the fitted  $R^2$  of per-layer linear maps, as auxiliary indicators to examine the correspondence across all layers, and

focus on functional stitching through reconstruction. On the universality of representations across families, Wang et al. [18] compared Transformers and Mamba. They extracted the features of both models with sparse autoencoders, which are trained to express the input as a combination of a small number of features, and reported that most features are similar. They further showed that the structure of induction circuits, which predict the next token by referring to repeated patterns in the context, is also similar. Their subject, however, is Mamba, not the RWKV family.

## 2.5 Model Merging and Knowledge Fusion

Methods that merge models in weight space include model soups [19], which average the weights of several models fine-tuned from the same pretrained model; TIES-Merging [20], which merges task-specific weight differences after resolving sign conflicts; and evolutionary model merge [21], which searches for merging recipes by evolutionary computation. These methods assume identical or compatible weights. FuseLLM [22], which fuses the output distributions of several models, does not need compatible weights, but it needs a procedure that aligns tokens between the vocabularies of different tokenizers and additional training of the target model. This study merges no weights and recombines frozen blocks by converting in activation space, which makes it complementary to these methods.

## 2.6 Position of This Study

Among the studies above, those that compose language models either connect two Transformers, fine-tune the composed model as in Manticore, or use at most one pretrained parent model. A stitching scheme with a single connection point, connections in both directions, and frozen parent models has already been demonstrated by Chen et al. [11] for pairs of Transformers within one model series that share a tokenizer, and this study does not claim the scheme itself as a new contribution. What this study addresses is the case of different families, which Chen et al. did not examine: we test whether the same stitching holds between a recurrent model based on linear attention and a Transformer. To our knowledge, no study has composed these two families by connecting them at a single point with both parent models and the LM heads frozen and evaluated the result as a generator. We also report how much the composition reduces the Transformer KV cache.

# 3. Method

## 3.1 Target Models

We use RWKV-4-Raven-7B and Tulu-Pythia-6.9b, instruction-following models tuned to respond according to user instructions<sup>\*2</sup>. Below, the former is denoted model  $A$  and the latter model  $B$ . These symbols are used in the notation for residuals in §3.2 and for read-outs in

<sup>\*2</sup> The experiment code, raw evaluation data, and reproduction procedures are available at <https://github.com/Katagiri-Hoshino-Lab/NinaXander>.

§3.4. Both models have 32 blocks and a hidden width of  $d=4096$ , and they share the tokenizer of GPT-NeoX-20B, a language model developed by the non-profit research group EleutherAI. The same input is therefore converted into the same token sequence, and the residuals at each position can be paired directly. Because the depths are equal, the  $j$ th blocks of the two models can be paired, and because the hidden widths are equal, the vectors at the connected layer have the same dimension. These three conditions were chosen deliberately to make the connection easier, and pairs of models whose tokenizers, depths, or hidden widths differ are outside the scope of this study.

The base models of both were pretrained on The Pile [23], an 825 GiB English corpus that EleutherAI built from 22 kinds of data, including academic papers, web documents, books, and source code. Although the base corpus is shared, the composition of the data and the prompt formats used for additional training on pairs of instructions and responses, called instruction tuning below, differ between the two models.

The RWKV architecture [24] replaces softmax dot-product attention with linear-time sequence mixing by the WKV operator, which averages the values of the tokens up to the current one with weights set by their keys and decayed over time, and it can run the same computation either in parallel form or in recurrent form. The Raven model used here is an instruction-following version of RWKV-4 developed by the RWKV project and released by its developer BlinkDL. RWKV-4-Pile-7B, pretrained on The Pile itself, is further trained on data that include Alpaca [25], the instruction-response data released by Stanford University, the instruction data Guanaco and GPT4All, Code-Alpaca, and ShareGPT. Code-Alpaca is synthetic data of about 20,000 instructions and responses centered on code generation, editing, and optimization [26]. ShareGPT collects multi-turn conversations with ChatGPT, the dialogue service of OpenAI, shared by its users [27]. We convert the RWKV-4-Raven-7B released by BlinkDL into the format used on the model-sharing platform Hugging Face and re-save it in 16-bit floating point (FP16)<sup>\*3</sup>. The weights we use have 7,392,649,216 parameters, 32 blocks, a hidden width of 4096, and an intermediate width of 16384 in channel mixing. The converted configuration sets `context_length` to 1024. This value is defined as the maximum sequence length handled in a single forward pass, and in recurrent form the model can propagate a state whose size does not depend on the context length<sup>\*4</sup>. The file name and revision of the source weight file, however, were not recorded, so the context length used to train the source checkpoint cannot be identified. Being recurrent also does not guarantee language performance on sequences longer than the training or configured length.

The Pythia series [28] consists of Transformer models that EleutherAI developed for research and trained with its GPT-NeoX library. Tulu-Pythia-6.9b is Pythia-6.9b with all of its parameters fine-tuned by the Allen Institute for AI (AI2), a non-profit research institute, on the mixed instruction data Tulu [29]. The mixture contains FLAN V2, which converts many natural language processing tasks into instruction format and whose name stands for Finetuned Language Net; Chain-of-Thought, which collects answers that include reasoning steps; Dolly, instruction data written by employees of Databricks; OpenAssistant 1, conversation data collected by crowdsourcing; GPT4-Alpaca, in which GPT-4, a large language model of OpenAI, responds to the Alpaca instructions; and Code-Alpaca and ShareGPT<sup>\*5</sup>. The weights we use have 6,856,040,448 parameters, 32 blocks, a hidden width of 4096, and 32 attention heads. The maximum number of positions in the configuration is 2048<sup>\*6</sup>. According to their publishers, the instruction-tuning data of both models include datasets named Code-Alpaca and ShareGPT, and Tulu’s GPT4-Alpaca uses the same instructions as Raven’s Alpaca. Raven’s publisher does not state versions, so the extent of the overlap cannot be determined, but the instruction data of the two models are not independent. In this paper, block and layer are used as synonyms.

### 3.2 Residuals and Notation

In this paper a residual does not mean an error; it is the hidden vector passed from each block to the next. Connecting two models means converting this vector from the coordinate system of one model into that of the other and passing it on. We write  $h_M^j[t] \in \mathbb{R}^{4096}$  for the unstandardized residual that block  $j$  of model  $M \in \{A, B\}$  outputs at token position  $t$ . It corresponds to  $h^\ell$  in §2, and the block index is  $j \in \{0, \dots, 31\}$ . We feed the same token sequence to both models and take  $h_A^j[t]$  and  $h_B^j[t]$  as one training example.

The scale of the residuals differs between models and layers, so we compute the mean and standard deviation of each dimension from the training data of each layer.  $\text{std}_M^j$  standardizes a residual with these statistics, and  $\text{unstd}_M^j$  is the inverse operation that restores the original scale. The standardized residual is written  $x_M^j[t] = \text{std}_M^j(h_M^j[t])$ . In the training objective below the layer index  $j$  and the token position  $t$  are omitted for brevity, and we write  $x_A, x_B$ .

### 3.3 Shared-Latent Adapter

Our adapter maps the intermediate representations of the two parent models into a common vector space, called the shared latent space below, and recovers the intermediate representation of the other model from this space. We call this adapter the shared-latent adapter. It has one encoder

<sup>\*3</sup> BlinkDL, RWKV-4 Raven model repository, <https://huggingface.co/BlinkDL/rwkv-4-raven>.

<sup>\*4</sup> Hugging Face Transformers, RWKV documentation, [https://huggingface.co/docs/transformers/model\\_doc/rwkv](https://huggingface.co/docs/transformers/model_doc/rwkv).

<sup>\*5</sup> Allen Institute for AI, open-instruct-pythia-6.9b-tulu model card, <https://huggingface.co/allenai/open-instruct-pythia-6.9b-tulu>.

<sup>\*6</sup> The released configuration file is <https://huggingface.co/allenai/open-instruct-pythia-6.9b-tulu/blob/main/config.json>, which sets `max_position_embeddings` to 2048.

and one decoder for each model.  $E_A, E_B$  are encoders that map standardized residuals into the shared latent space, and  $D_A, D_B$  are decoders that recover each model's standardized residuals from the shared latent space. The subscripts  $A, B$  indicate the model to which each map belongs.  $E_A, E_B, D_A, D_B$  form a single adapter shared by all 32 layers, with no conditioning on the layer index. Differences between layers are handled only by the statistics of standardization and unstandardization.  $z_A, z_B \in \mathbb{R}^{d_z}$  are the two latent vectors obtained at the same token position, and the latent width is  $d_z=4096$ . We call recovering a residual from the shared latent space a read-out. A read-out that uses the encoder and decoder of the same model is a self-reconstruction, and a read-out that combines the encoder of one model with the decoder of the other is a cross-family read-out. With  $\lambda \geq 0$  the hyperparameter that weights the latent alignment error against the two self-reconstruction errors, the training objective is defined as follows.

$$\begin{aligned} z_A &= \text{LN}(E_A(x_A)), & z_B &= \text{LN}(E_B(x_B)), \\ \mathcal{L} &= \|D_A(z_A + \xi_A) - x_A\|^2 + \|D_B(z_B + \xi_B) - x_B\|^2 \\ &\quad + \lambda \|z_A - z_B\|^2. \end{aligned} \quad (1)$$

The first and second terms of Eq. (1) are the self-reconstruction errors in restoring the residuals of  $A$  and  $B$  in the spaces of their own models. The third term is the latent alignment error, which pulls together  $z_A$  and  $z_B$  obtained at the same token position. The larger  $\lambda$  is, the larger the relative weight of the third term in the training loss  $\mathcal{L}$ . Each squared norm is computed as a mean squared error over all tokens and dimensions. LN is a non-affine layer normalization with no learnable scale or bias.  $\xi_A, \xi_B \sim \mathcal{N}(0, \sigma^2 I)$  are independent Gaussian noise terms with mean 0 added just before decoding.  $\sigma > 0$  is the hyperparameter that sets the standard deviation of the noise, and  $I$  is the  $d_z \times d_z$  identity matrix. In training the adapter we fix  $\lambda=1$  and  $\sigma=0.4$ . Setting  $\lambda=1$  gives the same coefficient 1 to the three mean-squared-error terms and does not mean that the terms take equal values during training. We did not tune these values per layer, direction, or task, nor run ablation experiments that vary them.

The loss contains no term that predicts the residual of  $B$  directly from that of  $A$ , nor one that predicts the residual of  $A$  from that of  $B$ . In a cross-family read-out,  $z_A$  obtained with the encoder of  $A$  is fed to the decoder of  $B$  to give  $D_B(z_A)$ , and the read-out from  $B$  to  $A$  uses  $D_A(z_B)$ . For these maps to work, the third term must bring  $z_A$  and  $z_B$  into the same coordinate system. Because the method uses paired residuals, we do not call it unsupervised learning, even though no cross-family target is given directly.

Each encoder and decoder consists of an input linear projection, one residual block, and an output linear projection. The residual block normalizes its input, applies a fully connected layer  $\text{fc}_1$ , the activation function GELU (Gaussian Error Linear Unit), and a fully connected layer  $\text{fc}_2$  in turn, and adds the result to its input, which is a pre-norm con-

figuration. Both the latent width and the output width of  $\text{fc}_1$  are 4096. Because the weights of  $\text{fc}_2$  are zero-initialized, the residual block starts as the identity map, and  $E$  and  $D$  begin training as linear maps. A non-affine LN lies between them, however, so a cross-family read-out as a whole is not linear even at initialization. Zero initialization is a property at the start of training and does not mean that the trained map is linear. Compared with the 14.25B parameters of the two parent models combined, the adapter has 268.5M parameters, a ratio of 1.9%.

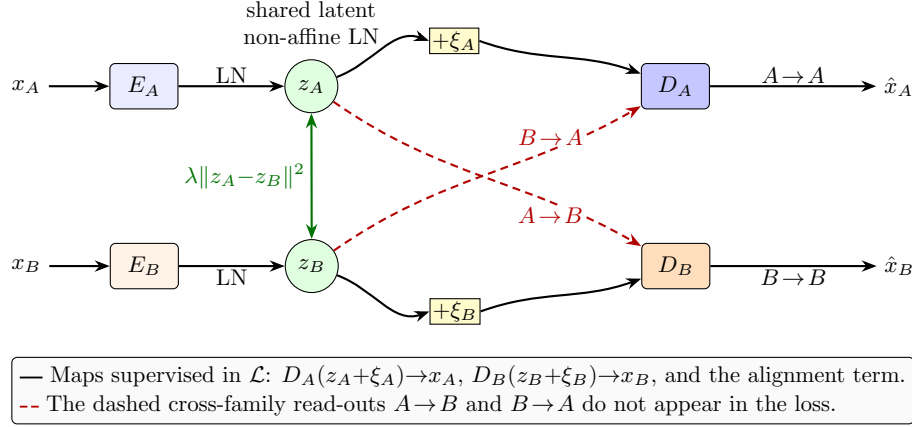
The noise  $\xi$  is introduced to keep the latent from being dominated by a constant component that does not depend on the token. We call this component the common mode. The alignment term also becomes small when  $z_A$  and  $z_B$  consist only of this constant component. Adding noise just before decoding, however, makes it hard to recover the original residuals from a latent with little per-token variation and thus raises the self-reconstruction error. The noise is a design that relies on this property to make the latent keep its per-token variation.

### 3.4 Configurations and Chimera Models

We call the layer at which the residual of the front model is converted by the adapter and passed to the back model the switch layer, the model that runs up to the switch layer the front model, and the model that runs after the switch layer the back model. A combination of a front and a back model is called a configuration, including the case in which the same model is used for both; a model with a specified switch layer and a parent model alone are also called configurations. The front-back combinations compared in this study are of four kinds: RWKV self-reconstruction control, RWKV→Pythia, Pythia self-reconstruction control, and Pythia→RWKV. RWKV→Pythia and Pythia→RWKV are cross-family chimera models, and RWKV self-reconstruction control and Pythia self-reconstruction control are self-reconstruction controls that pass through the same adapter at the same switch layer. The switch layer  $L$  is an index counted from 0 and denotes the last block that the front parent model runs. Because block 0 is counted, the front model runs  $L+1$  blocks. In the notation of §3.2, the conversion at the switch layer is given by the following equation, where a tilde marks a residual converted into the coordinate system of the other model.

$$\begin{aligned} \tilde{h}_B^L[t] &= \text{unstd}_B^L \left( D_B \left( \text{LN} \left( E_A(x_A^L[t]) \right) \right) \right), \\ \tilde{h}_A^L[t] &= \text{unstd}_A^L \left( D_A \left( \text{LN} \left( E_B(x_B^L[t]) \right) \right) \right). \end{aligned} \quad (2)$$

Block  $L+1$  of  $B$  receives  $\tilde{h}_B^L[t]$ , and block  $L+1$  of  $A$  receives  $\tilde{h}_A^L[t]$ . Equation (2) adds no training noise. Execution has three steps: running blocks 0 to  $L$  of the front parent model, converting the residual once, and running blocks  $L+1$  to 31 and the language-model head of the back parent model, as shown in **Fig. 2**. The conversion is applied only once, at the switch layer, with no round trip at each block. Every



**Fig. 1** Shared-latent adapter. Solid lines are the self-reconstructions and the alignment that the loss includes, and dashed lines are the cross-family read-outs that the loss does not include.

chimera model therefore runs 32 blocks in total,  $L+1$  in the front model and  $31-L$  in the back model. The error in the residual introduced by the conversion at the switch layer is called the conversion error. Which model serves as the front model is called the connection direction. In Pythia  $\rightarrow$  RWKV ( $L=4$ ), for example, the 5 blocks 0 to 4 of Pythia run first, followed by the 27 blocks 5 to 31 of RWKV.

A configuration is denoted by the names of its front and back models, as in RWKV  $\rightarrow$  Pythia, and the switch layer is added as in RWKV  $\rightarrow$  Pythia ( $L=4$ ). Read-outs, in contrast, are denoted by symbols such as  $A \rightarrow B$ . The read-out used at the switch layer of RWKV  $\rightarrow$  Pythia is  $A \rightarrow B$ , that used by Pythia  $\rightarrow$  RWKV is  $B \rightarrow A$ , and those used by RWKV self-reconstruction control and Pythia self-reconstruction control are  $A \rightarrow A$  and  $B \rightarrow B$ , respectively. In tables, RWKV is abbreviated as R and Pythia as P, and RWKV  $\rightarrow$  Pythia and RWKV self-reconstruction control are written, for example, as R  $\rightarrow$  P and R self.

### 3.5 Evaluation Metrics

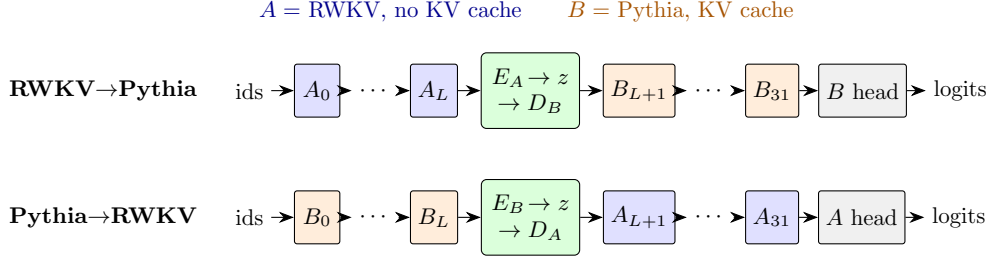
How well a residual is recovered is measured by the coefficient of determination  $R^2 = 1 - \text{SSE}/\text{SST}$ . SSE is the sum of squared errors between the predicted and true residuals, and SST is the total sum of squares of the true residuals after subtracting the per-dimension token mean. A value of 1 means perfect recovery, 0 means the same performance as always predicting the per-dimension mean, and a negative value means worse than this mean predictor. We measure four read-outs: the cross-family  $A \rightarrow B$  and  $B \rightarrow A$  and the self-reconstructions  $A \rightarrow A$  and  $B \rightarrow B$ . The latter two are controls that show the information lost by encoding and decoding. Below, the  $R^2$  of a read-out is called its read-out accuracy.

$R^2$  is computed per layer over the evaluation tokens and all 4096 dimensions together. Because SST is the total variation after removing the per-dimension token mean, dimensions with larger variance contribute more. The alignment of the latent spaces is measured by the centered correlation  $\rho_{\text{ctr}}$ , the Pearson correlation coefficient of  $z_A, z_B$  flat-

tened into one dimension after subtracting the token mean from each dimension. Values close to 1 mean that the two latents vary in the same direction, and 0 means no linear correlation. We further define the varying fraction, the fraction of the per-token varying component of the latent, as  $f = \|z - \bar{z}\|^2 / \|z\|^2$ , where  $\bar{z}$  is the per-dimension token mean within a layer and  $f$  is computed separately for  $A$  and  $B$  and averaged. Each metric is computed per layer, and we report the simple mean over the 32 layers.

There are two sets of evaluation data, both taken from positions that do not overlap the training sequences. The main values in the text are computed on sequences of 112 tokens, 1,000 sequences or 112,000 tokens in total. These data are also used to adjust the learning rate and to select the checkpoint (§3.6). The per-layer analysis and the shuffle control, which re-pairs the tokens at random, use 400 sequences of the same length, 44,800 tokens in total. Because the evaluation data differ, the 32-layer means of the per-layer analysis differ slightly from the main values; both are given in §4.1. The affine control for accuracy evaluation in §4.3 is fitted on 40,096 rows of standardized residuals taken from the part of Alpaca not used to train the adapter. Each row is the residual vector at one token position. These 40,096 rows and the sequences used for the linearity evaluation and the intervention in §4.3 are leading parts of the main evaluation data, taken from the same positions by the same procedure, and are contained in it. The fitted  $R^2$  of the all-layer correspondence expresses how well the representation of one layer can be predicted linearly from that of another. A linear map is fitted by ridge regression, that is, least squares with L2 regularization, to the 22,400 rows shared by each pair of layers, which come from 200 sequences of 112 tokens taken from the same positions as the main evaluation data. The same samples are used for fitting and evaluation, so for a 4096-dimensional linear map the value is easily inflated.

The language ability of the chimera models is evaluated in three ways. First, English prompts are given without modification, and generation examples are obtained by deterministic argmax decoding at temperature 0, which selects



**Fig. 2** Configurations of the RWKV→Pythia and Pythia→RWKV chimera models.

the most probable token at each step. Generation is limited to 40 new tokens and stops early only when the model outputs EOS (end-of-sequence), the special token that marks the end of a sequence. Second, the log-likelihood of each option of a multiple-choice question is computed, the most likely option is taken as the answer, and accuracy is measured. The question sets are ARC-Easy [30], the subset of the ARC (AI2 Reasoning Challenge) dataset of science exam questions for elementary and middle school students that excludes the hard questions both a retrieval-based and a word co-occurrence baseline answered incorrectly, and SciQ [31]. Below, ARC-Easy is abbreviated as ARC. Both mostly have 4 options, and random choice gives an accuracy of about 25%.

We use all 2376 test questions of ARC and all 1000 test questions of SciQ. SciQ is evaluated in closed-book form, without its support passages, so our SciQ values cannot be compared directly with published values obtained under the standard protocol that provides those passages. Two scores are computed from the log-likelihoods of the option tokens conditioned on the question, that is, of the tokens after the prompt: their sum and their average. Options differ in length and the sum tends to be smaller for longer options, so the text mainly uses the latter, the length-normalized score. Differences between configurations are computed from paired data on the same questions, so they are evaluated with McNemar’s exact test for paired binary data and with a paired bootstrap that estimates the variability of the difference by resampling pairs of the same questions. The text and tables give McNemar  $p$ -values, and bootstrap confidence intervals of the differences are included in the released data. Accuracy differences are expressed in percentage points, called points below.

Third, the next-token cross-entropy is measured on Alpaca and on WikiText-103 [32], a corpus built from good-quality Wikipedia articles and called WikiText below, to compare language-modeling performance inside and outside the domain of text used to train the adapter, called the training domain below.

### 3.6 Training Setup

The adapter is trained on the residuals of 2,075 sequences of 112 tokens taken from the beginning of the Alpaca training split. The distinct token positions number about 232k, and the residual pairs number 7.44M over the 32 layers. This

set of pairs was iterated about 228 times. For optimization we used AdamW, which decouples weight decay from Adam (Adaptive Moment Estimation), with an effective batch size of 4096, an initial learning rate of  $10^{-3}$ , and a minimum learning rate of  $10^{-5}$ . At each iteration one of the 32 layers is chosen uniformly at random, and an effective batch is drawn from the residual pairs of the chosen layer. Gradients are clipped so that their norm does not exceed 1.0. Every 1,000 iterations,  $A \rightarrow B$  was measured on the evaluation data (§3.5), and a reference value was updated only by an evaluation that exceeded it by more than a relative  $10^{-3}$ ; the learning rate was multiplied by 0.5 each time 4 consecutive evaluations brought no update. We ran 416,000 iterations with random seed 0 and used the checkpoint at iteration 415,000, where  $A \rightarrow B$  on the evaluation data was best, for all subsequent evaluations. The checkpoint was not reselected based on  $B \rightarrow A$  or on multiple-choice results. Details of the implementation and the released artifacts are given in Appendix A.1.

## 4. Results

We first examine whether the representations of corresponding layers align, and measure how well each model’s residuals can be recovered from the resulting representation. We then analyze the nonlinearity of the adapter, and finally evaluate the accuracy, inference memory, performance outside the training domain, and generation examples of the composed chimera models.

### 4.1 Shared Representation Between Corresponding Layers

The adapter trained on the residuals of both models shows a centered correlation of  $\rho_{\text{ctr}} = 0.901$  on text not used for training. In addition to this value, we take as evidence that a shared representation exists the fact that the cross-family read-out accuracy is 0.559 for  $A \rightarrow B$  and 0.590 for  $B \rightarrow A$  and that both fall sharply under the shuffle control described next. The self-reconstruction read-out accuracies,  $A \rightarrow A$  0.695 and  $B \rightarrow B$  0.711, indicate how much information is lost by encoding and decoding.

This shared representation is based on per-token correspondence rather than on overlapping distributions. When the row order of  $A$  is kept and the residual rows of  $B$  are repaired under a single random permutation, on the evaluation data for the per-layer analysis  $\rho_{\text{ctr}}$  falls as  $0.901 \rightarrow 0.000$ , and

the cross-family read-out accuracy falls as  $0.561 \rightarrow -0.592$  for  $A \rightarrow B$  and as  $0.592 \rightarrow -0.609$  for  $B \rightarrow A$ . The self-reconstruction read-out accuracies do not change, by construction. Neither cross-family read-out therefore succeeds without the correct token correspondence.

Same-index layer pairs, however, are not intrinsically special. As **Fig. 3** shows, comparing the unstandardized residuals  $h$  directly, without the adapter, over all  $32 \times 32$  layer pairs gives a small linear CKA overall. The mean of the diagonal elements, which are the pairs of layers with the same index, is 0.05, and the mean over the other pairs is 0.03. Layer  $j$  of  $A$  is most similar to layer  $j$  of  $B$  in only 2 of 32 rows. For 14 of the 32 layers of  $A$  the most similar layer of  $B$  is layer 0, and for 12 it is the last layer. Same-index layers are not special in linear predictability either. Fitting a linear map to every layer pair gives fitted  $R^2$  values of 0.62–0.94, high for almost every pair. These values are measured on the rows used for fitting, however, and cannot be compared with values evaluated on rows not used for fitting, such as those of the direct affine maps in §4.3. The mean fitted  $R^2$  is 0.87 on the diagonal and 0.84 off the diagonal, and the same-index layer is best in only 3 of 32 rows. The value  $\rho_{\text{ctr}} = 0.901$  is therefore an alignment obtained by training on the chosen pairs of same-index layers, not a correspondence inherent in layers that share an index. In other words, this study is a case study that connects same-index layer pairs in a setting where any layer pair can be predicted by linear regression to some degree.

The Pearson correlation coefficient of  $z_A, z_B$  before centering, called the raw correlation below, is  $\rho_{\text{raw}} = 0.965$ , which is higher than the centered correlation 0.901. The difference is explained by the common mode. With the varying fraction  $f$  defined in §3.5, the following holds under a layer normalization with no per-element coefficients when the varying fractions of  $A$  and  $B$  are equal and their common modes are parallel.

$$\rho_{\text{raw}} = (1 - f) + f \rho_{\text{ctr}}. \quad (3)$$

Here  $f$  is the mean of the varying fractions of  $A$  and  $B$ . At evaluations after training has progressed, Equation (3) holds to within about  $3 \times 10^{-4}$ . At iteration 0, however, the error is  $6.6 \times 10^{-2}$ . The checkpoint at iteration 415,000 has  $f = 0.350$ , so the token-independent common mode accounts for 65% of the squared norm of the latent. Equation (3) reproduces the raw correlation exactly as  $(1 - 0.350) + 0.350 \times 0.901 = 0.965$ , which shows that most of the raw correlation is due to the common mode, which contains no information. We therefore use the centered correlation as the measure of alignment.

Note that  $f$  is a second moment, not an amount of information.  $f \rightarrow 0$  means that the latent has degenerated to a constant and contains no per-token information, whereas a large  $f$  only shows that the latent varies in direction. Whether the varying component contains information is judged by the read-outs. The read-out measure is a centered  $R^2$ , which no token-independent predictor can raise

above 0. Hence  $B \rightarrow B = 0.711$  cannot be obtained from a constant prediction, which is consistent with the common mode accounting for 65% of the squared norm of the latent.

## 4.2 Self-Reconstruction Compared with Cross-Family Read-Outs

**Fig. 4** shows the relation among the five measures described in §4.1.  $\rho_{\text{ctr}}$  is a correlation coefficient and the other four are coefficients of determination, so they cannot be compared on the same scale. Among the cross-family read-out accuracies,  $B \rightarrow A$ , used by Pythia→RWKV, slightly exceeds  $A \rightarrow B$ , used by RWKV→Pythia, so the RWKV residuals are easier to recover from the latent obtained with the Pythia encoder. Both read-outs fall below the self-reconstruction that uses the same decoder and differs only in the input latent. For the read-out accuracies,  $A \rightarrow B \leq B \rightarrow B$  and  $B \rightarrow A \leq A \rightarrow A$  hold in all 32 layers. This ordering is observed in experiments rather than guaranteed by theory; still, in both directions substantial error already exists at the self-reconstruction stage, and improving the cross-family part does not remove the self-reconstruction error.

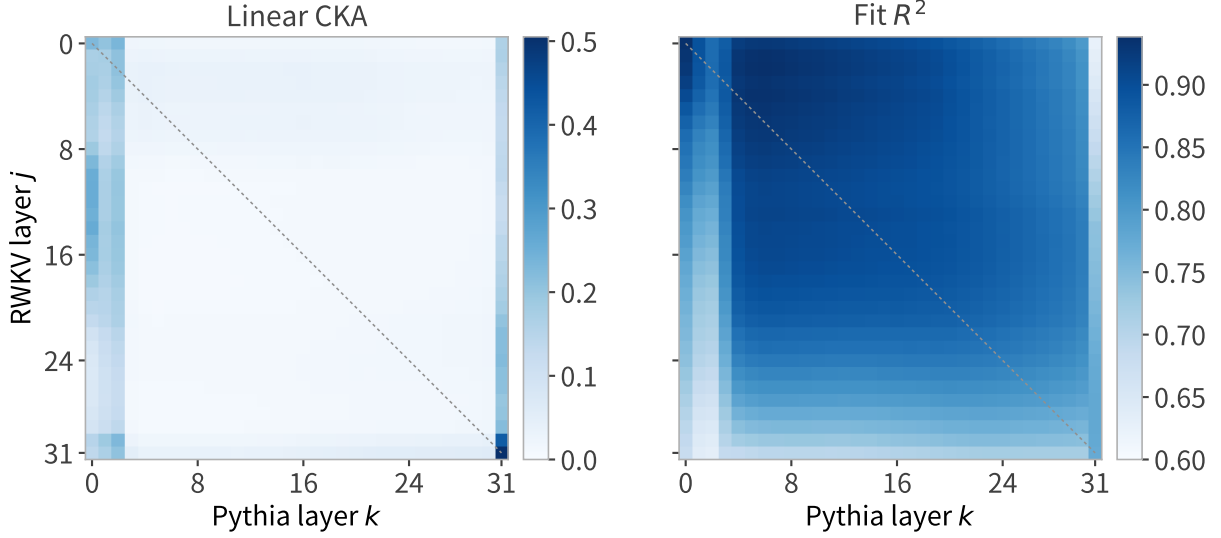
At  $d_z = d = 4096$  the latent is as wide as the residual, and the non-affine LN removes only two degrees of freedom per token, the mean and the norm. Latent width or rank is therefore unlikely to constrain reconstruction strongly. Even so,  $B \rightarrow B$  is only 0.711. Whether this is an upper bound set by the noise  $\sigma$  is unknown. We have also not isolated whether the error arises in the encoder, the latent normalization, the decoder, or the training noise, and what limits the reconstruction accuracy remains unresolved.

## 4.3 Nonlinearity of the Adapter

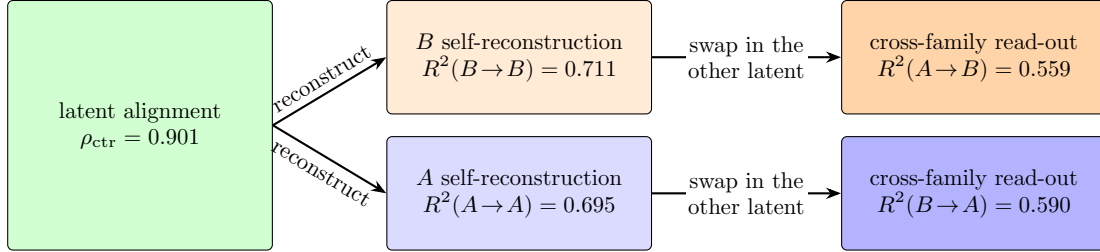
As described in §3.3,  $E$  and  $D$  of the adapter begin training as linear maps. This is a property before training, however, not after 415,000 iterations. We therefore examine the linearity of the adapter at this checkpoint with three affine maps that play different roles. Maps (i) and (ii) were measured for both RWKV→Pythia and Pythia→RWKV on the same 5 representative layers  $\{4, 10, 16, 22, 28\}$  and the same fitting and evaluation rows. The 5 representative layers were taken every 6 layers from layer 4 to layer 28. The rows are standardized residuals of 900 sequences of 112 tokens taken from positions of Alpaca not used to train the adapter; for each layer the first 50,400 rows are used for fitting and the last 50,400 rows for evaluation.

(i) The imitation affine map is the affine map that best approximates the output of the trained adapter, and it is used to measure how linear the adapter is. The fraction of the adapter’s output variance that it explains is 86.6% for RWKV→Pythia and 88.0% for Pythia→RWKV. The remaining 13.4% and 12.0% therefore cannot be explained on the evaluation rows by the affine map fitted on the first half of the rows, and the trained map is nonlinear in both directions.

(ii) The direct affine map predicts, without the adapter,



**Fig. 3** Correspondence over all  $32 \times 32$  layer pairs of the unstandardized residuals  $h$  taken without the adapter. Left: linear CKA. Right: fitted  $R^2$  of per-layer linear maps. Dashed lines mark same-index layers, and the two panels use different color scales.



**Fig. 4** Relation between the shared latent space and the four read-outs. Read-outs of the same color share a decoder.

the standardized output residual from the standardized input residual for each layer and direction; it is the best affine map fitted by ridge regression. The regularization coefficient is  $10^{-3}$  times  $\text{tr}(X^\top X)/(d+1)$  for the input matrix  $X$  with an intercept column appended. This map measures how much cross-family prediction a low-capacity map achieves. On the evaluation rows, the best direct affine map gives 0.487 for RWKV $\rightarrow$ Pythia and 0.535 for Pythia $\rightarrow$ RWKV. On the same rows the adapter gives 0.558 for RWKV $\rightarrow$ Pythia and 0.576 for Pythia $\rightarrow$ RWKV, so the nonlinear map adds +0.071 and +0.041. These values were evaluated on the 5 representative layers, and both the layers and the rows differ from those of the 32-layer means 0.559 and 0.590 in §4.1. That even a low-capacity affine map predicts about half of the variance of the other model’s residual in both directions, 87% and 93% of the adapter’s values, is consistent with the conclusion that a shared representation exists.

(iii) The affine control for accuracy evaluation embeds an affine map like (ii) in the chimera model and serves as a control for comparing multiple-choice accuracy with the adapter. It was fitted for each switch layer on the rows described in §3.5, by ridge regression with an intercept and the same regularization coefficient as (ii). As **Table 1** shows, replacing the adapter of RWKV $\rightarrow$ Pythia with this control lowers accuracy by 8–23 points. For Pythia $\rightarrow$ RWKV, by

**Table 1** Length-normalized accuracy (%) of the adapter (Ad) and of the affine control for accuracy evaluation (Af). R $\rightarrow$ P: RWKV $\rightarrow$ Pythia; P $\rightarrow$ R: Pythia $\rightarrow$ RWKV.

| $L$ | ARC               |      |                   |      | SciQ              |      |                   |      |
|-----|-------------------|------|-------------------|------|-------------------|------|-------------------|------|
|     | R $\rightarrow$ P |      | P $\rightarrow$ R |      | R $\rightarrow$ P |      | P $\rightarrow$ R |      |
|     | Ad                | Af   | Ad                | Af   | Ad                | Af   | Ad                | Af   |
| 4   | 59.6              | 41.9 | 62.2              | 60.2 | 70.9              | 47.5 | 69.2              | 69.6 |
| 8   | 59.0              | 43.1 | 57.3              | 56.0 | 67.7              | 49.1 | 64.6              | 68.0 |
| 16  | 48.6              | 34.8 | 57.5              | 57.5 | 54.5              | 35.9 | 68.0              | 69.6 |
| 24  | 47.0              | 39.4 | 59.6              | 59.2 | 51.1              | 42.7 | 69.9              | 68.2 |

contrast, the difference is small and lies within  $-3.4$ – $+2.0$  points over the 8 conditions. At  $L=4$  on ARC the adapter is 2.0 points higher ( $p=0.005$ ), whereas at  $L=8$  on SciQ the affine control is 3.4 points higher ( $p=0.003$ ). The two maps differ in training objective, data, and regularization, so we do not treat this difference between directions as a causal effect of nonlinearity.

Next, we intervene to disable the nonlinearity of the adapter. To suppress the confounds above, in each residual block of the same trained adapter we scale by a coefficient  $\alpha$  only the output of the branch that normalizes the input and applies  $\text{fc}_1$ , GELU, and  $\text{fc}_2$ . This branch is the only learned nonlinearity in the adapter, and we call its output the nonlinear part.  $\alpha=1$  is the original map, and  $\alpha=0$  is the

**Table 2** Change in length-normalized accuracy (%) when the nonlinear part is disabled,  $\alpha : 1 \rightarrow 0$ . R→P: RWKV→Pythia; P→R: Pythia→RWKV.

| $L$ | ARC       |           | SciQ      |           |
|-----|-----------|-----------|-----------|-----------|
|     | R→P       | P→R       | R→P       | P→R       |
| 4   | 59.6→31.9 | 62.2→31.8 | 70.9→31.3 | 69.2→32.6 |
| 8   | 59.0→29.9 | 57.3→35.5 | 67.7→31.4 | 64.6→37.0 |
| 16  | 48.6→28.5 | 57.5→43.7 | 54.5→31.7 | 68.0→47.3 |
| 24  | 47.0→34.8 | 59.6→48.4 | 51.1→40.0 | 69.9→49.1 |

**Table 3** Length-normalized accuracy (%) of the four configurations. The maximum of each row and task is in bold. R self: RWKV self-reconstruction control; P self: Pythia self-reconstruction control; R→P: RWKV→Pythia; P→R: Pythia→RWKV.

| $L$ | R self      |      | R→P  |      | P self      |             | P→R         |      |
|-----|-------------|------|------|------|-------------|-------------|-------------|------|
|     | ARC         | SciQ | ARC  | SciQ | ARC         | SciQ        | ARC         | SciQ |
| 4   | <b>62.8</b> | 68.5 | 59.6 | 70.9 | 62.2        | <b>74.8</b> | 62.2        | 69.2 |
| 8   | 59.8        | 66.7 | 59.0 | 67.7 | <b>62.7</b> | <b>74.8</b> | 57.3        | 64.6 |
| 16  | 54.8        | 59.8 | 48.6 | 54.5 | 57.4        | <b>71.1</b> | <b>57.5</b> | 68.0 |
| 24  | 52.2        | 57.9 | 47.0 | 51.1 | 59.4        | <b>70.1</b> | <b>59.6</b> | 69.9 |

same map with the nonlinear part removed. Sweeping  $\alpha = 0, .25, .5, .75, 1$  for the mean cross-family read-out accuracy of the 5 representative layers on the evaluation data for the per-layer analysis (§3.5) gives  $-0.41, -0.75, 0.35, 0.52, 0.56$  for RWKV→Pythia and  $-0.30, -0.66, 0.36, 0.53, 0.58$  for Pythia→RWKV. Removing the nonlinear part lowers the read-out accuracy to negative values in both directions, and the dependence on  $\alpha$  is non-monotonic. Even at  $\alpha=0$  the non-affine LN remains, so the map does not become entirely linear. The input and output linear projections of the encoder and the decoder were trained jointly with the nonlinear part, and in this trained adapter the nonlinear part is indispensable to both the RWKV→Pythia and Pythia→RWKV read-outs.

The effect of the nonlinear part in **Table 2** is  $+11.1$ – $+39.6$  points for RWKV→Pythia and  $+11.2$ – $+36.6$  points for Pythia→RWKV, and McNemar’s exact test on the same questions gives  $p < 2 \times 10^{-9}$  in all 16 comparisons. Removing the nonlinear part from this trained adapter therefore lowers the task accuracy as well as the read-out accuracy for both RWKV→Pythia and Pythia→RWKV. This is, however, a property of removing the nonlinear part from a jointly trained map, and it does not show that nonlinearity is generally necessary for connecting families. We did not compare against a linear adapter trained for the same objective, and the accuracy difference between the affine control and the adapter is small for Pythia→RWKV (Table 1).

#### 4.4 Accuracy Compared with the Parent Models

This subsection compares the chimera models with the two frozen parent models to examine how much accuracy is retained relative to the stronger parent and whether the weaker parent is exceeded. The evaluation method is as described in §3.5.

**Table 3** gives the accuracies of the four configurations,

**Table 4** Accuracy differences (points) between the main configurations and McNemar exact-test  $p$ -values in parentheses. R→P: RWKV→Pythia; P→R: Pythia→RWKV.

| $L$ | ARC                      |                         |                          | SciQ                     |                 |                          |
|-----|--------------------------|-------------------------|--------------------------|--------------------------|-----------------|--------------------------|
|     | vs RWKV                  |                         | Dir.                     | vs RWKV                  |                 | Dir.                     |
|     | R→P                      | P→R                     | P→R<br>–R→P              | R→P                      | P→R             | P→R<br>–R→P              |
| 4   | –2.2<br>(0.042)          | +0.4<br>(0.66)          | +2.6<br>(0.015)          | +3.9<br>(0.018)          | +2.2<br>(0.14)  | –1.7<br>(0.28)           |
| 8   | –2.8<br>(0.007)          | –4.5<br>( $< 10^{-4}$ ) | –1.7<br>(0.11)           | +0.7<br>(0.69)           | –2.4<br>(0.081) | –3.1<br>(0.052)          |
| 16  | –13.2<br>( $< 10^{-4}$ ) | –4.3<br>( $< 10^{-4}$ ) | +8.9<br>( $< 10^{-4}$ )  | –12.5<br>( $< 10^{-4}$ ) | +1.0<br>(0.50)  | +13.5<br>( $< 10^{-4}$ ) |
| 24  | –14.8<br>( $< 10^{-4}$ ) | –2.2<br>(0.043)         | +12.6<br>( $< 10^{-4}$ ) | –15.9<br>( $< 10^{-4}$ ) | +2.9<br>(0.082) | +18.8<br>( $< 10^{-4}$ ) |

and **Table 4** gives paired tests for the main comparisons. In Table 4, “vs RWKV” is the accuracy difference between each configuration and RWKV alone, and “Dir.” is the difference between Pythia→RWKV and RWKV→Pythia at the same switch layer; both are based on the length-normalized score. The comparisons with Pythia alone are omitted from the table. The main results are the following five points.

First, even though the front and back models come from different families, both RWKV→Pythia and Pythia→RWKV answer the questions, and every configuration exceeds the accuracy of random choice, about 25%, by a wide margin. Second, the chimera models are significantly less accurate than the stronger parent Pythia, whose accuracy is ARC 65.3% and SciQ 81.0% (**Table 5**), in all 16 conditions over both directions, all switch layers, and both tasks ( $p \leq 0.003$ ). Third, among the chimera models, the only one that significantly exceeds RWKV, whose accuracy is ARC 61.8% and SciQ 67.0%, is RWKV→Pythia ( $L=4$ ) on SciQ, and Pythia→RWKV does not significantly exceed RWKV at any switch layer. The advantage of RWKV→Pythia ( $L=4$ ) appears only with the length-normalized score; with the summed likelihood score, the differences from RWKV alone are  $-2.7$  points on SciQ for RWKV→Pythia ( $L=4$ ), and  $-4.6$  points on ARC and  $-5.7$  points on SciQ for Pythia→RWKV ( $L=4$ ). Fourth, the effect of the switch depth depends on the direction. RWKV→Pythia loses accuracy monotonically as the switch layer deepens, whereas Pythia→RWKV is non-monotonic and substantially exceeds RWKV→Pythia at the same switch layer for  $L=16, 24$ . On the evaluation data for the per-layer analysis, the cross-family read-out accuracies at the same four layers are  $0.570/0.555/0.575/0.567$  for RWKV→Pythia and  $0.552/0.517/0.604/0.608$  for Pythia→RWKV, so the accuracy ranking cannot be predicted from read-out accuracy alone. Fifth, the loss from encoding and decoding also appears in the accuracy of the self-reconstruction controls: the difference from Pythia alone is significant for Pythia self-reconstruction control at every switch layer on both tasks, and the difference from RWKV alone is significant for RWKV self-reconstruction control on both

**Table 5** Accuracy and inference memory. KV@4k: KV cache size at context length 4096. P alone: Pythia; R alone: RWKV; R→P: RWKV→Pythia; P→R: Pythia→RWKV.

| Config. | $L$ | Trans-<br>former<br>layers | KV/token<br>(KiB) | KV@4k<br>(GiB) | Reduc.<br>(%) | ARC/<br>SciQ |
|---------|-----|----------------------------|-------------------|----------------|---------------|--------------|
| P alone | –   | 32                         | 512               | 2.00           | 0             | 65.3/81.0    |
| R→P     | 4   | 27                         | 432               | 1.69           | 15.6          | 59.6/70.9    |
| R→P     | 8   | 23                         | 368               | 1.44           | 28.1          | 59.0/67.7    |
| R→P     | 16  | 15                         | 240               | 0.94           | 53.1          | 48.6/54.5    |
| R→P     | 24  | 7                          | 112               | 0.44           | 78.1          | 47.0/51.1    |
| P→R     | 24  | 25                         | 400               | 1.56           | 21.9          | 59.6/69.9    |
| P→R     | 16  | 17                         | 272               | 1.06           | 46.9          | 57.5/68.0    |
| P→R     | 8   | 9                          | 144               | 0.56           | 71.9          | 57.3/64.6    |
| P→R     | 4   | 5                          | 80                | 0.31           | 84.4          | 62.2/69.2    |
| R alone | –   | 0                          | 0                 | 0              | 100           | 61.8/67.0    |

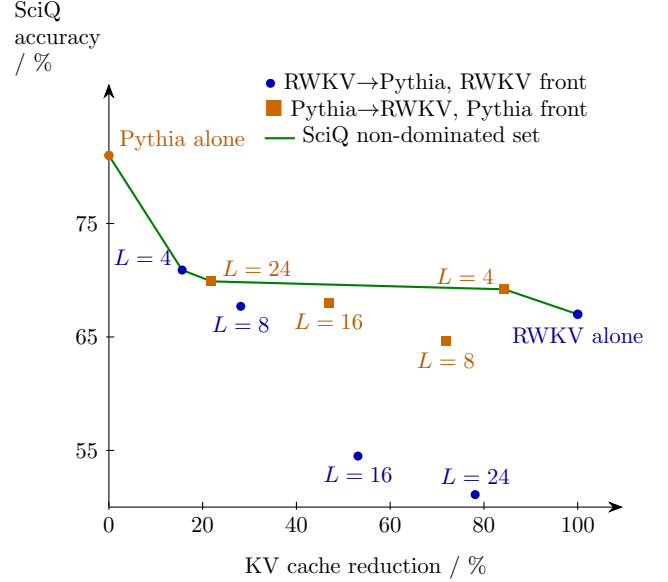
tasks at  $L=16, 24$ . At  $L=8$  it is significant only on ARC ( $p=0.006$ ). The differences from the self-reconstruction controls, however, are not due to conversion error alone. With Pythia as the back model, accuracy decreases in the order of Pythia alone, Pythia self-reconstruction control, and RWKV→Pythia, and the difference between Pythia self-reconstruction control and RWKV→Pythia is 3–12 points on ARC and 4–19 points on SciQ. With RWKV as the back model, in contrast, Pythia→RWKV exceeds RWKV self-reconstruction control at deep switch layers, by +2.7/+7.4 points on ARC and +8.2/+12.0 points on SciQ at  $L=16/24$ . A difference from a cross-family configuration also contains the difference in ability between the front parent models, so it cannot be read as the loss caused by conversion.

#### 4.5 KV Cache Reduction

For each layer and token, Pythia’s KV cache holds two 4096-dimensional vectors, the key and the value, in FP16, totaling 16 KiB, or 512 KiB over 32 layers. RWKV has no KV cache but instead keeps a recurrent state of 64 KiB per layer that does not depend on the context length. This value is computed from the reference implementation, which keeps 2 of its 5 state vectors in FP16 and 3 in 32-bit floating point (FP32). The KV cache sizes in Table 5 exclude the recurrent state. RWKV→Pythia keeps  $31-L$  Transformer layers, and Pythia→RWKV keeps  $L+1$ . The reduction applies to the KV cache, not to the resident weights. The adapter itself occupies 0.50 GiB in FP16 and about 0.25 GiB even when only one direction is kept in memory, which is comparable to the reduced KV cache. Even when the unused blocks are physically removed, the resident weights of a chimera model, including the adapter, are 13.49–14.55 GiB, about the same as the 13.77 GiB of RWKV alone and the 12.77 GiB of Pythia alone. The output of the implementation with the blocks removed matched that of the implementation without removal.

Comparing RWKV→Pythia and Pythia→RWKV, Pythia→RWKV ( $L=4$ ) reduces the Transformer layers from 27 → 5 relative to RWKV→Pythia ( $L=4$ ) while keeping the two-task mean accuracy at an equivalent level,

65.25% → 65.70%. Because the ranking depends on the task, **Fig. 5** shows the trade-off between KV cache reduction and accuracy on SciQ. We call the non-dominated set the set of configurations for which no other configuration is at least as good in both measures and better in at least one of them.



**Fig. 5** SciQ accuracy and KV cache reduction of RWKV→Pythia and Pythia→RWKV. The green line connects the configurations in the non-dominated set.

Adding the configurations with Pythia as the front model yields Pythia→RWKV ( $L=4$ ), which has a high reduction rate and cannot be obtained from the configurations with RWKV as the front model alone. RWKV alone, on the other hand, needs no KV cache, and its accuracy does not differ significantly from that of Pythia→RWKV ( $L=4$ ) (§4.4). The advantage of the method is therefore that it creates new trade-off points between frozen parent models, not that new configurations uniformly exceed the parent models.

If the only goal is to reduce the KV cache, the same reduction is also obtained by running only the first blocks of Pythia and connecting their output to the final layer normalization and the LM head. We call this method early exit. We therefore compared the accuracy of each chimera model with that of an early exit that keeps the same number of Transformer blocks (**Table 6**). In the 5 configurations that reduce the Transformer layers to 17 or fewer, the chimera model significantly exceeds the early exit on both tasks. Pythia→RWKV ( $L=4$ ) scores 62.2% against 26.8% on ARC and 69.2% against 30.5% on SciQ, compared with the early exit that stops after the first 5 blocks of Pythia. In the 3 configurations that keep 23 or more layers, by contrast, the differences are small; only ARC for RWKV→Pythia ( $L=8$ ), which keeps 23 layers, is significant, and RWKV→Pythia ( $L=4$ ), which keeps 27 layers, does not differ from the early exit. However, because the early exit runs fewer blocks and keeps fewer resident weights at the same KV cache size, this comparison does not equalize computation.

**Table 6** Differences in length-normalized accuracy (points, chimera model minus early exit) against a Pythia early exit that keeps the same number of Transformer layers, with McNemar exact-test  $p$ -values. Transformer: number of Transformer layers kept. R→P: RWKV→Pythia; P→R: Pythia→RWKV.

| Config. | $L$ | Trans-<br>former | ARC   |             | SciQ  |             |
|---------|-----|------------------|-------|-------------|-------|-------------|
|         |     |                  | Diff. | $p$         | Diff. | $p$         |
| P→R     | 4   | 5                | +35.4 | $< 10^{-4}$ | +38.7 | $< 10^{-4}$ |
| R→P     | 24  | 7                | +18.5 | $< 10^{-4}$ | +18.7 | $< 10^{-4}$ |
| P→R     | 8   | 9                | +27.6 | $< 10^{-4}$ | +31.6 | $< 10^{-4}$ |
| R→P     | 16  | 15               | +11.2 | $< 10^{-4}$ | +12.8 | $< 10^{-4}$ |
| P→R     | 16  | 17               | +16.9 | $< 10^{-4}$ | +17.9 | $< 10^{-4}$ |
| R→P     | 8   | 23               | +4.9  | $< 10^{-4}$ | +1.6  | 0.36        |
| P→R     | 24  | 25               | +2.1  | 0.069       | +0.5  | 0.80        |
| R→P     | 4   | 27               | -0.2  | 0.87        | -0.9  | 0.59        |

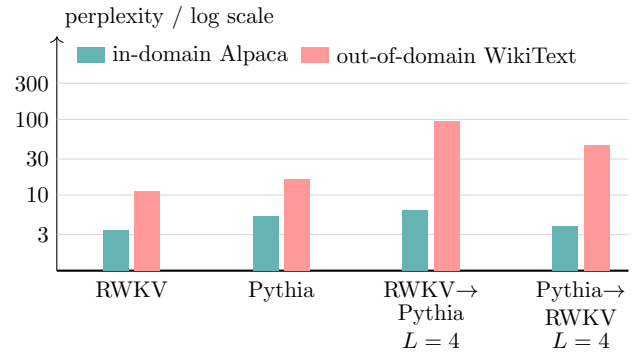
As for inference time, with the implementation in the Hugging Face Transformers library used in this study and the two parent models in FP16 placed on 2 NVIDIA V100 GPUs, we compared the time to process the whole input sequence at once, taking the median of 3 measurements. At context length 512, RWKV alone took 6,538 ms and Pythia alone 99 ms, a factor of about 66. This gap arises because the implementation used runs the RWKV recurrence sequentially, token by token. At context length 2048 the times were 25,933 ms and 385 ms, a factor of about 67. For chimera models with the unused layers removed, configurations of the same direction take longer the more RWKV blocks they run; at context length 512, RWKV→Pythia ( $L=4$ ), which runs 5 RWKV blocks, took 1,128 ms, and Pythia→RWKV ( $L=4$ ), which runs 27 RWKV blocks, took 5,652 ms. These values depend on the implementation used for the measurement and do not compare the computational cost of the architectures.

#### 4.6 Effect of Domain Shift

This subsection examines how performance changes when the evaluated text comes from a domain different from the training domain, a mismatch called domain shift below. The accuracies above were measured on short questions close to instruction format. Because a smaller KV cache matters for long contexts, we vary the context length and measure language-modeling performance on Alpaca, inside the training domain, and on WikiText, outside it. As the measure we use perplexity, the exponential of the next-token cross-entropy, which is better when smaller. **Fig. 6** shows the results at context length 512. The perplexities of the parent models are 3.42 on Alpaca and 11.33 on WikiText for RWKV, and 5.19 and 16.22 for Pythia. Alpaca, however, overlaps the instruction data of both parents, so the parents' Alpaca perplexities may partly reflect exposure during instruction tuning. RWKV→Pythia ( $L=4$ ) gives 6.40 on Alpaca and 93.87 on WikiText, and Pythia→RWKV ( $L=4$ ) gives 3.85 and 45.00; Pythia→RWKV ( $L=4$ ) is lower than RWKV→Pythia ( $L=4$ ) in both domains, but on WikiText it is well above those of the parent models. Extend-

ing the context length to 2048 gives 3.50 on Alpaca and 34.78 on WikiText for Pythia→RWKV ( $L=4$ ), so performance does not degrade and even improves. The main cause of the degradation is therefore domain shift, not context length. Pythia→RWKV is less affected by domain shift than RWKV→Pythia, but the WikiText perplexity of 34.78 at context length 2048 is 4.2 times the 8.27 of RWKV, the parent model with the lower perplexity at the same context length, and remains high.

We also compared the adapter with a control that places at the switch layer an affine map fitted by ridge regression with an intercept, as in (iii) of §4.3. This control is re-fitted for each switch layer and each evaluated domain on its first 39,936 tokens, which are not used for evaluation; it is thus fitted even to WikiText, on which the adapter was not trained, so this comparison is unfavorable to the adapter. For Pythia→RWKV, the WikiText perplexity of the affine control is lower than that of the adapter at every switch layer and context length; for Pythia→RWKV ( $L=4$ ) it is 21.73 against 45.00 at context length 512 and 15.74 against 34.78 at context length 2048. On Alpaca the two are nearly equal, 3.86 against 3.85 for Pythia→RWKV ( $L=4$ ) at context length 512. For RWKV→Pythia, in contrast, the WikiText perplexity of the affine control is at least that of the adapter at every switch layer and context length. With this caveat, the result that the affine control is better for Pythia→RWKV suggests that part of the degradation outside the training domain may arise from the conversion by the adapter, which was trained only on Alpaca.



**Fig. 6** Perplexity on Alpaca and WikiText at context length 512. The vertical axis is logarithmic.

#### 4.7 Generation Examples of the Chimera Models

The adapter of the checkpoint at iteration 415,000 was inserted at  $L=4$  in both RWKV→Pythia and Pythia→RWKV, and responses to the same four English prompts were generated by argmax decoding under the conditions of §3.5. Two of the prompts use instruction format, which separates the instruction and the response with the headings “### Instruction:” and “### Response:”, and two use completion format, which asks the model to continue a sentence.  $L=4$  is the switch layer at which the two-task mean accuracy is highest among the four switch layers in each direction, and it is the common switch layer at which

both directions belong to the non-dominated set in the plane of KV cache reduction and accuracy on SciQ (Fig. 5). This choice, however, was made after seeing the results of the evaluation tasks.

These examples show that both RWKV→Pythia and Pythia→RWKV can generate syntactically well-formed output. Both directions answer the capital of France correctly and begin listing the primary colors correctly. RWKV→Pythia also answers the black-hole instruction correctly in one sentence, so generation that follows the instruction is achieved. On the other hand, content or instruction format can be lost even when fluency is kept. Both directions answer “hydrogen” wrongly in the completion about the composition of water, and Pythia→RWKV drifts into a dialogue after answering the primary colors correctly and explains the black hole circularly. Current chimera models can therefore lack factual accuracy. RWKV alone, however, also gives the wrong answer about water and drifts after the primary colors, and a causal relation between the conversion error at  $L=4$  and individual wrong answers has not been verified.

## 5. Discussion

In this experiment, a shared representation with  $\rho_{ctr} = 0.901$  could be learned between corresponding layers of frozen models from different families. The range in which this conclusion applies, however, is narrow. The correspondence was measured under the favorable conditions of the same tokenizer, hidden width, depth, and input, in a domain close to Alpaca. The shuffle control rules out a token-independent match of distributions, but we have not separated whether the observed correspondence arises from meaning, vocabulary, position, or syntax. Performance also decreases sharply on WikiText, outside the training domain (§4.6). This study therefore claims only the existence of a shared representation that corresponds token by token under favorable conditions, not a general common semantic space.

A successful functional stitch does not mean that the two models represent the same information. Smith et al. [33] showed that stitching can succeed between different tasks, between images and bird-song audio, and even from randomly generated, clustered pseudo-representations. The stitching layers of their discriminative models, however, were trained with the task loss, the scheme that Balogh and Jelasy [34] point out rates distant layers as similar by producing out-of-distribution representations. Our adapter, which uses no task loss and is trained with self-reconstruction and latent alignment, is closer to the direct matching of representations that the same paper finds to balance structural and functional requirements. Smith et al. nonetheless obtained apparent alignment even with low-capacity stitching layers and with autoencoder stitches trained by matching latent codes, and our adapter has higher capacity. Our main reason for judging that the observed alignment cannot be explained by the capacity of the adapter alone is that a low-capacity affine map already achieves substantial cross-family

prediction (§4.3).

A high latent correlation also does not guarantee high read-out accuracy, so alignment and decodability must be evaluated separately, and other work further distinguishes decodability from use by the receiving model. The preprint of Zhang and Xin [35] showed that a linear map that transfers activations between small models of the Pythia series achieves  $R^2 = 0.883$  in a normalized space, yet replacing the hidden states of the receiving model with the converted activations lowers performance on tasks that require several reasoning steps, even after rescaling to the receiver’s norm, and weak additive injection does not improve it. They concluded that the training objective should optimize how the receiving model uses the activations; this point also applies to our adapter, whose objective does not include the receiving model’s output. The unstandardization in Equation (2) differs from their norm rescaling in restoring the per-dimension mean and scale, but we have not isolated the contribution of this difference.

## 6. Limitations of This Study

Some of the evaluation questions may be contained in the training data of a parent model. Pythia was instruction-tuned on a subset of FLAN V2, and the tasks that make up FLAN V2 include ARC and SciQ. Matching the corresponding part of the released mixture verbatim finds 19 ARC questions and 4 SciQ questions among the evaluation questions. The matching covers only verbatim matches and cannot detect paraphrases or formatting differences. Questions shorter than 6 words, which we excluded to avoid chance matches, were not searched either, 40 for ARC and 11 for SciQ. The detected counts are therefore lower bounds. The matching also covers only the FLAN V2 and Chain-of-Thought parts of Pythia’s mixture, not RWKV’s instruction data or The Pile, on which both models were pretrained. Re-computing the accuracy of every configuration from the per-question correctness records without these questions changes it by at most 0.22 points and does not change the conclusion of any comparison discussed in the text. On these questions, moreover, the accuracy of RWKV, whose listed instruction data do not include FLAN V2, changes relative to the remaining questions in the same direction as that of Pythia, by  $-9.2$  points on ARC and  $+8.0$  points on SciQ, so this pattern is also consistent with an explanation based on the difficulty of the questions themselves. ARC and SciQ nonetheless remain kinds of tasks that Pythia has already learned, so the values in this paper should be read as measuring the retention of ability after composition rather than the acquisition of new ability.

The adapter was trained with a single random seed, and although 2 tasks×2 directions×4 switch layers and several controls were evaluated, no correction for multiple comparisons was applied. The main read-out accuracies were also computed on evaluation data that were used to adjust the learning rate and to select the checkpoint, and may be somewhat optimistic. The same caution applies to the values in

**Table 7** Generation examples of the two parent models and of RWKV→Pythia and Pythia→RWKV at  $L=4$ .

| Prompt (verbatim)                                                                  | Pythia alone                                                                                                          | RWKV→Pythia ( $L=4$ )                                                                                               | RWKV alone                                                                                                              | Pythia→RWKV ( $L=4$ )                                                                                                       |
|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| ### Instruction:<br>What is the capital of France?<br>### Response:                | “The capital of France is Paris.”                                                                                     | “The capital of France is Paris.” ✓                                                                                 | “Paris.”                                                                                                                | “Paris.” ✓                                                                                                                  |
| The three primary colors are                                                       | “red, yellow, and blue. These colors are often used in combination to create a wide variety of colors.”               | “red, yellow, and blue.” ✓                                                                                          | “red, blue, and yellow.” then drifts into a Q&A asking for green, the opposite color of red                             | “red, blue, and yellow.” then drifts into a Bob/Alice dialogue about green                                                  |
| Question: Water is made of hydrogen and<br>Answer:                                 | “Hydrogen is made of hydrogen and oxygen.”                                                                            | “Hydrogen.” wrong                                                                                                   | “hydrogen.” wrong                                                                                                       | “hydrogen.” wrong                                                                                                           |
| ### Instruction:<br>Explain what a black hole is in one sentence.<br>### Response: | “A black hole is a region of spacetime where gravity is so strong that nothing, not even light, can escape its pull.” | “A black hole is a region in space where gravity is so strong that nothing, not even light, can escape its pull.” ✓ | “A black hole is a region of space where the gravitational pull is so strong that nothing, not even light, can escape.” | “A black hole is a type of celestial object that is formed when a star collapses and collapses into a black hole.” circular |

§4.3, which use sequences overlapping these evaluation data. The intervention on the nonlinear part was performed on a single checkpoint, and whether it generalizes to adapters trained under different conditions is unknown. Each composition also has a single switch layer, and how conversion error accumulates over several switch layers has not been measured. Furthermore, the multiple-choice evaluation is based on likelihood and does not directly measure the ability to generate the correct answer, and the four generation examples in each direction are qualitative supplements, not a comprehensive evaluation of generation ability.

The long-context cross-entropy was measured up to context length 2048 to match the maximum number of positions in Pythia’s configuration. Measurement points beyond the 1024 specified by the converted RWKV configuration lie outside the range the configuration assumes, and, as noted in §3, the context length used to train the original checkpoint cannot be confirmed. This evaluation also measures language-modeling quality and does not directly measure reading comprehension that requires long-range dependencies. Ablations over  $\sigma$ ,  $\lambda$ , the latent width, the number of residual blocks, and the amount of training data, as well as an alignment control with two models of the same family, were not carried out in this study and are left for future work.

## 7. Conclusion

We built NinaXander, a series of chimera models that connect frozen blocks from different architecture families at a single point through one shared-latent adapter, and tested with RWKV and Pythia whether frozen models from different families can be recombined post hoc. Under favorable conditions the alignment between corresponding layers is learnable, and the 8 chimera models over 2 directions  $\times$  4 switch layers, obtained from one adapter without retraining, answered multiple-choice questions, and the 2 configurations

at  $L=4$  produced syntactically well-formed text. Pythia→RWKV ( $L=4$ ) reduced the Transformer KV cache by 84.4% with accuracy not significantly different from that of RWKV alone, but it did not exceed Pythia. The configurations that exceed a Pythia early exit with the same KV cache size by a wide margin on both tasks, however, are those that reduce the Transformer layers to 17 or fewer. In all layers, each cross-family read-out was also lower than the self-reconstruction that uses the same decoder, and performance decreased sharply on WikiText, outside the training domain. The significance of NinaXander therefore lies not in a higher-performing new model but in showing a way to recombine frozen models post hoc. The contribution of this study is to show that this recombination is partly possible and can create new trade-off points between frozen parent models, and to describe, on the basis of measurements, the possibilities and limits of composition in one favorable case. Future work includes generalization through training across several domains, identifying what limits the self-reconstruction accuracy, selecting complementary parent models, measuring how conversion error accumulates over several switch layers, and reproduction with multiple random seeds.

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- <https://huggingface.co/Katagiri-Hoshino-Lab/ninaxander-tulu69-to-raven7b-best>
- RWKV→Pythia with the default  $L=4$ :  
<https://huggingface.co/Katagiri-Hoshino-Lab/ninaxander-raven7b-to-tulu69-best>

The latter two bundle the two frozen parent models in FP16, the shared tokenizer, the adapter, and custom inference code that selects the switch layer from  $L=4, 8, 16, 24$ , so each runs as a standalone chimera model. Each is fixed to one connection direction, and its default switch layer  $L=4$  was chosen after seeing the evaluation results, as in §4.7. No independent data split was used for this selection. All three repositories contain model cards in English, Japanese, and Simplified Chinese, a NOTICE file that records the attribution of the parent models, weights in the Safetensors format, and a file list with SHA-256 hashes. The license is the AI2 AI Model License for non-commercial research, matching Pythia, the more restrictive parent model.

## Appendix

### A.1 Reproduction Information

- Models: the two parent models of §3 are used in FP16. The released artifact `parent_fingerprints.json` records the SHA-256 (Secure Hash Algorithm 256-bit) hashes of the local parent models used for training.
- Adapter training: the adapter was trained on 4 NVIDIA V100 GPUs.
- Data: the evaluation sequences were taken from Alpaca from character position 2,000,000 onward, which does not depend on the number of training tokens. The standardization statistics are computed from the training residuals of each layer and are also used for unstandardization.
- Evaluation: for the RWKV back model of Pythia→RWKV, options are always evaluated with a batch size of 1.
- Code and structured data: the repository given in §3 contains the experiment code, the raw data of each evaluation with aggregate tables of the reported values, the reproduction procedures, and a check script that automatically matches the numbers in the text against the aggregate tables.
- Hugging Face releases: the inference artifacts as of July 30, 2026 are stored in the following three model repositories.
  - Shared adapter: <https://huggingface.co/Katagiri-Hoshino-Lab/ninaxander-raven7b-tulu69-adapter>
  - Pythia→RWKV with the default  $L=4$ :